School of Computer Science and Artificial Intelligence

WEEK-12 Assignment # 23

Program : B. Tech (CSE)

Specialization :AIML

Course Title : Competitive Programming

Course Code : 23CS302PC305

Semester : VI

Academic Session : 2025-2026

Name of Student : B.Thanmai

Enrollment No. : 2303A52126

Batch No. : 33

1.Subset Generation Using Bitmask

Elaborated Scenario:

In combinatorics and computer science, generating all subsets of a set is a common problem.

Suppose a company wants to evaluate all possible combinations of features for a product. Each feature can be either included or excluded. Representing subsets using bit masking allows efficient generation of all possible subsets without recursion. Task:

- Given a set of n elements, generate all possible subsets.
- Use bit masking where each bit represents whether an element is included (1) or excluded (0).
- Print the subsets in lexicographical or input order.

Example Input:

A B C

Expected Output:

{}

{A}

{B}

{C}

{A, B}

{A, C} {B, C}

{A, B, C}

Explanation:

There are 2^n = 8 possible subsets for a set of 3 elements. Each subset corresponds to a bitmask of length n:

000 → {}

001 → {C} 010 → {B}

011 → {B, C} 100 → {A}

101 → {A, C} 110 → {A, B}

111 → {A, B, C}

Python:

cp.py > ...

```python
# Step 1: Input
n = int(input())
arr = input().split()

# Step 2: Total subsets = 2^n
total = 1 << n   # same as 2^n

# Step 3: Loop through all bitmasks
for mask in range(total):
    subset = []

    # Step 4: Check each bit
    for j in range(n):
        if (mask >> j) & 1:
            subset.append(arr[j])

    # Step 5: Print subset
    if len(subset) == 0:
        print("{}")
    else:
        print("{" + ", ".join(subset) + "}")
```

```
3
a b c
{}
{a}
Downloads/hpc_log_app_fixed/cp.py
3
a b c
{}
{a}
{b}
{a, b}
3
a b c
{}
{a}
{b}
{a, b}
a b c
{}
{a}
{b}
{a, b}
{}
{a}
{b}
{a, b}
{b}
{a, b}
```

Java:

cp.java

```java
import java.util.*;

public class SubsetBitmask {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Step 1: Input
        int n = sc.nextInt();
        String[] arr = new String[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.next();
        }

        // Step 2: Total subsets
        int total = 1 << n; // 2^n

        // Step 3: Loop through all bitmasks
        for (int mask = 0; mask < total; mask++) {

            System.out.print("{");
            boolean first = true;

            // Step 4: Check each bit
            for (int j = 0; j < n; j++) {
                if ((mask & (1 << j)) != 0) {

                    if (!first) {
                        System.out.print(", ");
                    }

                    System.out.print(arr[j]);
```

cp.java

```java
public class SubsetBitmask {
    public static void main(String[] args) {
        int total = 1 << n; // 2^n

        // Step 3: Loop through all bitmasks
        for (int mask = 0; mask < total; mask++) {

            System.out.print("{");
            boolean first = true;

            // Step 4: Check each bit
            for (int j = 0; j < n; j++) {
                if ((mask & (1 << j)) != 0) {

                    if (!first) {
                        System.out.print(", ");
                    }

                    System.out.print(arr[j]);
                    first = false;
                }
            }

            System.out.println("}");
        }

        sc.close();
    }
}
```